Signed Primitive Annuli and Compressed Basis Certificates for Unknot Hierarchies

Exact simultaneous recognition in rank-two support blocks, source-authenticated implementation, and a research programme

ProveIt project — research draft
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Abstract

We develop and implement a signed terminal rule for the compressed normal-cut pipeline in ProveIt. After a geometrically certified core cancellation, we retain the ordered, signed conjugacy classes of every pending annulus. We then certify that these classes are simultaneously part of one free basis. Gordon's primitive-loop theorem converts this algebraic certificate into disjoint geometric dual disks, yielding an exact handlebody census.

The main algorithmic theorem is an exact polynomial-time decision procedure for compressed cyclic families whose generator-support components have rank at most two. In rank two, one lift of the inverse exponent-sum matrix decides a pair: apply the lift once and test for two distinct generator conjugacy classes. We give a self-contained subgroup-heredity proof for the block reduction, a linear shared-grammar construction for signed transport through column-private cancellations, and a binary-power form of backward Nielsen replay. Classical ingredients and derived consequences are distinguished.

The implementation adds separately written source, quotient, and basis checkers, with opt-in integration before the established whole-cut fallbacks. Five native normal cuts that the baseline refuses acquire independently verified certificates; Regina agrees on all five, each giving two genus-two handlebodies. An independent literal Whitehead audit finds no disagreement on 8,486 inputs. Compressed stress tests reach words longer than 2^{2048} without materializing them.

These results improve a genuine local decision stage. They do not establish the complete quasi-polynomial unknot recognizer. We state the missing global obligations against the announced target $2^{O((\log n)^3)}$, derive a conditional complexity-transfer statement, and propose concrete further questions on marked geometry, higher-rank residuals, and hierarchy search.

Keywords. Unknot recognition; compressed normal surfaces; handlebody; primitive multicurve; straight-line program; Nielsen automorphism; certificate verification; hierarchy complexity.

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1 Research outcome and the asymptotic target

1.1 What this contribution changes

The useful improvement is a change in what the terminal operator proves. The existing operator can certify a pending annulus by finding a private cocore in an already selected geometric disk system. Its unsigned crossing counts are sufficient witnesses, but they can retain backtracking and conjugating segments that disappear in the fundamental group. They also describe only that disk system. A primitive family may require another one.

The new operator first authenticates the actual boundary curves, including their traversal signs. It transports their conjugacy classes through the certified core reduction, and asks whether there exists one common free basis containing the resulting classes. This changes a fixed-disk-system test into an exact algebraic test on a useful restricted class. Classical topology then supplies the simultaneous dual disks. The algebra is applied only after the core is known geometrically to be a handlebody.

The distinction between *each curve is primitive* and *the whole family is primitive* is essential. Parallel copies of a primitive curve pass every separate single-curve test but cannot have distinct simultaneous dual disks. The common automorphism and distinct target letters in our certificate enforce the family condition directly.

Claim	Established here	Precise scope
Signed quotient transport	Linear number of grammar productions	Complete row-leaf or column-private source schedules; polynomial bit overhead
Common basis replay	Polynomial certificate verification	A supplied powered Nielsen sequence, in any rank
Exact discovery	Deterministic polynomial algorithm	Cyclic support blocks of at most two generators
Geometric terminal	Exact handlebody census	Authenticated core and all pending annuli; disk interfaces allowed
Native improvement	Five new certified cuts	Fixed, embedded report-57 fixtures; independent replay and Regina agreement
Global target	Conditional accounting only	No bound proved here for complete hierarchy discovery or repair

Table 1: Claim ledger. Grammar size, verification, discovery, and the global recognizer are different statements.

1.2 Repository baseline and relation to prior work

All comparisons use repository revision `e0b2f802f336fcb51e9ec5f29174cfe28dfb9d39`, whose commit message is "New research reports." The inspected sources include `Topology/UnknotRecognition`, its maintained synthesis, and the incoming reports present at that revision [8]. In particular, the existing column-peeling theorem already decides *complete positive row cancellation in a fixed unsigned incidence matrix.* Reordering the same positive pivots cannot defeat its stopping set. That theorem and its sparse timestamp checker are prior ProveIt work, not results of this article.

The baseline also contains actual core-side annulus transport, a joint bundle boundary kernel, adaptive lower forests, certified top 2–3 cancellation, and deterministic compressed

word algorithms. The incoming reports on exceptional rows, matroidal dual forests, nonlinear activation forests, and ordered interface kernels were reviewed before choosing this direction. A fresh search among those already covered unsigned choices would risk reproducing existing improvements. The present extension instead retains signs and cyclic order, then changes the free basis in a verified handlebody. Its native fixtures are checked against the complete baseline wrapper, including its adaptive forests and boundary-kernel fallback.

For integration, the patch preserves `annulus_method='counts'` as the default and adds `annulus_method='signed_rank_two'`. This permits direct comparison and preserves existing certificate formats. The experimental option is conservative: unsupported blocks, failed primitive tests, and exhausted resource allowances yield a refusal by this method. They never produce a negative topological verdict.

1.3 The published and announced complexity statements

Lackenby's March 2021 slides state the target, for a diagram with n crossings, as

$$T(n) = 2^{O((\log n)^3)} = n^{O((\log n)^2)}. \tag{1}$$

The slides describe a polynomial bound on surface complexity and an $O((\log n)^2)$ hierarchy-length bound, using efficient encoding, multisurfaces, and Heegaard/Cheeger-region arguments [6]. The target is not the stronger expression $n^{O(\log n)}$.

The July 2026 paper gives a new hierarchy algorithm and explicitly leaves possible speed-up to future work. Its Proposition 9.1 bounds iterations by $L(g+1)^L$, where L bounds hierarchy length and g bounds pattern complexity. Its Theorem 14.4 already gives polynomial-time compressed cutting data for uniform-type handle structures, in the handle count and the logarithm of extended surface weight [7]. We therefore treat compressed cutting as an established mathematical ingredient, while assessing the particular executable ProveIt contracts separately.

The remaining problem is quantitative and constructive: maintain adequate marked geometric information, choose useful surfaces, control discarded suffixes after repairs, and close every terminal state required by a complete recognizer. Faster arithmetic in one cut is useful only if those outer operations satisfy compatible bounds. Section 8 makes this accounting explicit.

1.4 Classical ingredients and the contribution's novelty

The single-curve relation between primitive elements and dual disks, and Gordon's theorem on primitive sets of boundary loops, are classical [1, 2, 11]. The rank-two fact $\mathrm{Out}(F_2) \cong \mathrm{GL}_2(\mathbb{Z})$ is also classical. Kapovich's July 2026 treatment supplies the compressed single-class rank-two algorithm and the needed uniqueness of primitive conjugacy classes with a prescribed homology vector [4]. Shared evaluation of supplied automorphism sequences and compressed free reduction belong to the established theory [10].

Accordingly, the pair criterion below is presented as a derived simultaneous corollary, not a new classification of rank-two primitive elements. The contribution is the complete chain from source-oriented annuli through compressed quotient transport, one common basis certificate, a correct geometric terminal, independent replay, and new native successes. The support-block reduction and explicit source-size accounting make the restricted algorithm precise enough to use in a larger complexity proof.

2 Input model, signs, and proof obligations

2.1 Three sizes that must remain separate

Let n be the crossing count of an original diagram, t the number of tetrahedra in a supplied triangulation, and W the total number of normal disks represented by its normal coordinates. The normal coordinates are binary integers. A local normal-cut input size includes t, coordinate bit lengths, and the combinatorial boundary data; it does not include W as an explicitly expanded list.

For the signed terminal, let L be the total number of signed source passages in real and passive words after forest contraction, r the total number of surviving core generators, p the number of cancelled real rows, and q the number of pending annuli. Let S be a shared binary straight-line program's number of productions plus its root-list size, and let B be the total binary size of the supplied powered Nielsen moves. Generator and node identifiers also consume bits. These quantities refer to an authenticated compressed cut. An estimate in S is not automatically an estimate in n.

Definition 2.1 (Shared word grammar)**.** A straight-line program, or SLP, is a finite acyclic directed graph with terminal rules $A \to x_j^{\pm 1}$ and concatenation rules $A \to BC$, together with a list of roots. The empty word is permitted as a distinguished root. Different roots may share descendants. Its structural size counts rules and roots; its encoded size also counts references and labels.

A binary grammar with S rules can represent a word of length exponential in S. Storing the length at every node can already use quadratic total bits. Thus an $O(S)$ production count does not claim linear bit time or linear bytes. We use exact integers and deterministic compressed word operations; no probabilistic equality test is used to decide a certificate.

2.2 Authenticated geometric state

The enclosing ProveIt operator starts from a valid triangulation and admissible normal coordinates under its existing compact orientable one-torus source contract. It authenticates the compressed parallelity decomposition and the boundary attachment skeleton. In the supported state:

1. Each remaining core component has a geometric cancellation certificate proving it is an orientable handlebody, with a specified free marking.
2. Every bundle base, including every isolated base, has nonempty boundary. Each nondisk bundle with a core interface has multiplicity one in this source representation. Its orientable total space is a handlebody. Isolated bundle multiplicities remain binary counts.
3. Each pending circle is the midpoint section of an actual annular interface, with an authenticated core owner and base owner. The circles are embedded, pairwise disjoint, and disjoint from real attaching regions.
4. Every disk-base bundle absorbed into the core contributes the real two-handle specified by its independently checked directed source path.
5. All surviving disk interfaces and every pending annulus are accounted for in the final incidence graph and source Euler identity.

These are premises supplied by the existing source machinery and replayed by the enclosing verifier. A JSON list of free-group words does not establish them. In particular, a presentation that simplifies algebraically to a free group is not used as a substitute for a handlebody certificate.

2.3 Recovering signed words without losing loop directions

Choose the authenticated forest in the core's zero/one-handle graph. Orient each nonforest one-handle and use the signed letter $\pm(h+1)$ for crossing handle h. Delete forest letters only after the path's geometric direction has been recovered. The resulting words are the usual graph-based free marking before the real two-handles are attached.

A crucial implementation detail is that a loop one-handle has identical endpoint owner IDs. Comparing the start and stop *owners* cannot tell which direction it was traversed. Instead the compiler compares ordered tetrahedron/face flags with the one-handle's source and target flags. This applies both to annular midpoint paths and to original real attaching cycles. The source checker reconstructs these flags independently.

After forest contraction, different choices of a path to the basepoint change a peripheral word by conjugation. Reversing its orientation changes it to its inverse. Both are allowed by the primitive-family definition. Erasing signs, reordering letters, or freely reducing an unauthenticated source path is a different operation and cannot supply the same evidence.

2.4 What is preserved by top 2–3 cancellation

The optional second signed schema reuses the existing geometric cancellation of a two-handle against a three-handle. In this supported operation, all zero- and one-handles are retained and each surviving two-handle keeps its source cycle, with an authenticated renumbering. Hence the annular path charts and the signed one-handle marking remain available. The verifier first replays this reduction and only then reconstructs the signed source packet in the reduced core.

This is a narrow statement about the established reduction. It does not assert that arbitrary handle moves preserve source charts automatically, and it supplies no general procedure for transporting a complete marked boundary pattern through future disk-system changes.

3 Primitive annuli from authenticated compressed multiwords

The input to this section is a geometrically authenticated handlebody core, not an arbitrary presentation of a free group. Its remaining annular midpoint curves are embedded and pairwise disjoint in the boundary. Their words must be obtained from the signed source paths and the certified core cancellations. The role of algebra is to find a different disk system for these already authenticated curves.

3.1 A classical geometric bridge and its relative form

An ordered family $(w_1, \dots, w_q)$ in a free group F_r is *conjugate primitive* if there is one automorphism Φ of F_r, distinct basis letters $x_{j_1}, \dots, x_{j_q}$, and signs $\epsilon_i \in \{-1, 1\}$ such that $\Phi(w_i)$ is conjugate to $x_{j_i}^{\epsilon_i}$ for every i. The conjugators may differ; the automorphism may not.

Proposition 3.1 (Classical bridge, relative to disk ports)**.** *Let H be an orientable handlebody of genus r, and let $\beta_1, \dots, \beta_q$ be disjoint embedded simple closed curves in ∂H. Their conjugacy*

classes form a conjugate primitive family if and only if there are disjoint properly embedded disks $D_1, \dots, D_q$ *with*

$$|\partial D_i \cap \beta_j| = \delta_{ij}.$$

The disks may be chosen to avoid any prescribed finite family of disjoint disk attachment regions in ∂H *that is disjoint from the curves. Consequently sufficiently thin annular neighborhoods of the curves form a simultaneous primitive family relative to those disk regions.*

Proof. We give the reduction to classical results to make the exact hypothesis clear. For one simple boundary curve, being a primitive element of the free fundamental group is equivalent to possessing a dual meridian disk [3, Section 4]. In particular, attaching a two-handle along it gives a handlebody of genus one less. These are the usual primitive-curve facts.

Suppose the classes form a conjugate primitive family. For any subset $J \subseteq \{1, \dots, q\}$, attach the two-handles indexed by J in any order. After some of these attachments, the fundamental group is the quotient by the normal closure of the corresponding basis elements. The remaining indexed curves are conjugates of distinct remaining basis elements, and hence are primitive. They remain embedded boundary curves because the original curves were disjoint. The one-curve result applies at every stage, so every subset of the two-handle attachments gives a handlebody. Gordon's theorem on primitive sets of boundary loops [2] now supplies simultaneous dual disks. This use of Gordon's theorem is essential: requiring only the manifold obtained after *all* attachments to be a handlebody is a weaker condition in general.

Conversely the dual disks have connected complement: in the graph dual to cutting along them, the curve β_i crosses exactly the edge indexed by D_i exactly once, so that edge is a loop. The dual graph is connected and all its edges are loops, hence it has one vertex. Extend the dual disks to a complete disk system. In its free basis, the word of β_i contains its own distinguished generator once and all the other distinguished generators zero times. It has the form $U_i x_{j_i}^{\epsilon_i} V_i$, where U_i, V_i use only the remaining generators. Simultaneous Nielsen changes fixing those remaining generators send these words to distinct basis letters, up to conjugacy.

Finally isotope the boundaries of the dual disks off the prescribed disk regions. An outermost arc of intersection in one of those regions can be pushed to its boundary collar; choose outermost arcs for the whole disjoint boundary family. This removes intersections without changing intersections with the β_i, since the disk regions and their sufficiently small collars avoid those curves. No essential meridian boundary can be contained entirely in a disk region. Isotopy extension carries this simultaneous boundary isotopy to the properly embedded disks. After taking thin annular neighborhoods, each unique crossing becomes one spanning arc. □

This proposition is an application of established topology, not a new classification theorem. It does not assert that the disks avoid additional *annular* ports whose curves were left out of the multiword. Such a relative annular assertion needs an additional theorem or certificate. Our terminal rule therefore includes every pending annulus in the relevant core component.

3.2 Linear grammar size for signed passive transport

Proposition 3.2 (Signed triangular source transport)**.** *Suppose a geometrically verified core has a complete column-private cancellation schedule* $(r_1, g_1), \dots, (r_p, g_p)$*. Here column-private means that, when* r_i *is deleted,* g_i *occurs once in* r_i *and zero times in every other remaining real attaching word, counting unsigned geometric passages. Let* L *be the total length of the original signed real attaching words and the signed passive words after deletion of source-forest letters. There is a shared straight-line program of size* $O(L + r + p + q)$ *representing all final passive*

conjugacy classes over the r surviving generators. It can be constructed without expanding any substituted word.

Proof. Every later real attaching word contains none of the earlier pivot generators. Thus the donor words are unchanged source words; there is no positive fill among the real rows. Write the donor at stage i as

$$W_{r_i} = A_i g_i^{\epsilon_i} B_i, \qquad \epsilon_i \in \{-1, 1\}.$$

Its relation solves to

$$g_i = (B_i A_i)^{-\epsilon_i}.$$

Every generator on the right is either a later pivot or a survivor. Process the pivots in reverse order, replacing each such generator occurrence by a reference to its already constructed root. The inverse of a root is a shared formal inverse, or can be implemented by a second grammar with reversed products at a constant-factor increase in size. Each original letter contributes only a bounded number of grammar symbols. Translate the passive source words once using these roots.

The defining equations are triangular. Eliminating them is an explicit Tietze isomorphism from the source core presentation to the free group on the surviving generators. Since the core's handlebody status and source paths were authenticated separately, the resulting roots represent the actual passive curves in a free basis of that handlebody. Binarization preserves the stated linear bound. In particular every expanded word has length at most $2^{O(L+r+p+q)}$, whose binary length is polynomial. □

For the row-leaf shortcut, the donor at each step contains its pivot once and only previously killed generators otherwise. Induction sends every cancelled generator to the identity; the passive roots are obtained by one shared deletion map. This gives the same linear structural bound.

The complete column-peeling result already in the repository allows this schedule whenever its fixed positive cancellation matrix can be completely cancelled. We do not claim the same linear grammar construction for an arbitrary positive-fill schedule. Its new output here is the signed ordered multiword; the existing positive transport theorem supplies numerical incidence counts instead.

3.3 Replay of a prescribed powered Nielsen certificate

Use elementary Nielsen transformations consisting of inversions, swaps, and

$$x_i \mapsto x_i x_j^k \quad \text{or} \quad x_i \mapsto x_j^k x_i, \qquad i \neq j, \quad k \in \mathbb{Z},$$

with all other generators fixed. Exponents are binary integers.

Proposition 3.3 (Shared backward replay)**.** *Let a shared input grammar have size S and let a supplied sequence be $\psi_1, \ldots, \psi_m$, applied in that chronological order, so the resulting automorphism is $\Phi = \psi_m \circ \cdots \circ \psi_1$. Put*

$$B = \sum_{\text{powered steps}} (1 + \lceil \log_2(|k| + 1) \rceil) + \#\{\text{swaps and inversions}\}.$$

All transformed input roots have one shared grammar of size $O(S + r + B)$. The construction takes polynomial bit time; lengths have $O(S + B + 1)$ bits, apart from the encoding of generator labels. Whether this supplied sequence carries the input roots to the specified distinct generator conjugacy classes can then be verified using deterministic compressed free and cyclic reduction [10].

Proof. Scan the sequence backwards. Maintain roots $I_t = S_0(x_t)$ for the suffix automorphism S_0 already processed. When prepending $\psi : x_i \mapsto x_i x_j^k$, the new composite is $S_0 \circ \psi$, so the only changed root is

$$I_i^{\text{new}} = I_i^{\text{old}} (I_j^{\text{old}})^k.$$

For the left version use $(I_j^{\text{old}})^k I_i^{\text{old}}$. An inversion replaces one root by its inverse; a swap exchanges two old roots simultaneously. Binary powering adds $O(\log(|k|+1))$ grammar nodes. All newly appended nodes refer only to old nodes, so the construction is acyclic. Rebuild the original grammar once with its terminals mapped to the final I_t; the original sharing is retained across every passive root.

This proves the size bound and the composition convention. At each step the maximum expanded generator-image length increases by at most $1+|k|$. Combining this product with the original grammar's exponential length bound gives the stated bit bound. Apply compressed cyclic reduction to each transformed root and check that the results are distinct single basis letters, with the permitted signs. □

This is a source-specific use and binary-power formulation of the standard compressed evaluation of prescribed automorphism sequences. The general linear sharing principle is already present in Schleimer's work and in Kapovich's Proposition 3.2 [4]. It does not bound the length of a sequence that a search must discover.

4 An exact rank-two simultaneous terminal

Theorem 4.1 (One homology lift decides a pair in rank two)**.** *Let $u, v \in F(a, b)$ be given by straight-line programs, and let*

$$A = \begin{pmatrix} \exp_a(u) & \exp_a(v) \\ \exp_b(u) & \exp_b(v) \end{pmatrix}.$$

If $\det A \notin \{-1, 1\}$*, the two conjugacy classes are not a conjugate primitive pair. Otherwise choose* any *automorphism Φ whose abelianization matrix is A^{-1}. Then the pair is conjugate primitive if and only if*

$$\operatorname{cycred}(\Phi(u)) = a, \qquad \operatorname{cycred}(\Phi(v)) = b.$$

There is a deterministic polynomial-time algorithm on the compressed input that constructs a powered Nielsen lift and performs these two tests.

Proof. Necessity of the determinant condition follows by abelianization. If the tuple is conjugate primitive, compose its witnessing automorphism with a signed permutation so that its two classes go to $[a]$ and $[b]$ in the given order. Call the resulting automorphism Ψ. Its matrix must be A^{-1}.

The classical rank-two isomorphism $\mathrm{Out}(F_2) \cong \mathrm{GL}_2(\mathbb{Z})$ says that two automorphisms with the same abelianization differ by an inner automorphism; see the recalled proof of the classical rank-two facts in [4, Proposition 8.1]. Hence Φ and Ψ have the same action on conjugacy classes. The two displayed tests follow. Conversely, if those tests pass, Φ itself witnesses conjugate primitivity.

Exponent sums are obtained by addition and sign reversal on the grammar DAG; their bit lengths are polynomial in its size. The Euclidean algorithm expresses the unimodular matrix A^{-1} as a polynomial-size sequence of integer shears, row interchanges, and sign changes. Lift

these operations to powered Nielsen transformations. The previous proposition constructs their shared compressed action in polynomial size. Compressed cyclic reduction completes the decision in polynomial time. □

One explicit matrix procedure first sends the first column of A to $(1,0)^{\mathsf{T}}$ by Euclidean row operations. The new matrix has the form $\left(\begin{smallmatrix} 1 & t \\ 0 & \delta \end{smallmatrix}\right)$ with $\delta = \pm 1$. Invert the second row if needed, then subtract t times that row from the first. The total left multiplier is A^{-1}. The shear $\left(\begin{smallmatrix} 1 & k \\ 0 & 1 \end{smallmatrix}\right)$ is lifted, for example, by $b \mapsto ba^k$, and the other shear by $a \mapsto ab^k$. Keeping this column-vector convention explicit prevents a transpose or composition-order error.

For a single class, the corresponding compressed rank-two algorithm is already proved by Kapovich [4, Theorem 8.7]: test coprime exponent sums, construct a Euclidean lift sending that vector to $(1,0)^{\mathsf{T}}$, and test final cyclic length one. The underlying rank-two fact can also be seen directly. If a primitive element z has exponent vector $(1,0)^{\mathsf{T}}$, choose Ψ with $\Psi(a) = z$. Its matrix has the form $\left(\begin{smallmatrix} 1 & t \\ 0 & \delta \end{smallmatrix}\right)$, where $\delta = \pm 1$. The automorphism fixing a and sending b to $a^t b^\delta$ has the same matrix. The two automorphisms differ by an inner automorphism, so z is conjugate to a. Thus the single-class final test is also exact for any chosen homology lift.

The pair theorem is a short simultaneous corollary of the classical rank-two outer-automorphism fact. Its contribution here is an exact native annular-family terminal with a checkable common lift, rather than a claim to a new general primitivity algorithm.

4.1 Independent small generator blocks in arbitrary rank

Lemma 4.2 (Conjugate primitive families descend to containing subgroups)**.** *Let A be a subgroup of a free group F, and let a finite tuple of elements of A be conjugate primitive in F. It is conjugate primitive in A. Here the same basis-extension definition is used if A has infinite rank.*

Proof. We use the standard spanning-tree construction of a free basis [5]. Choose an ambient basis containing $z_1, \ldots, z_q$ such that each given element has the form $h_i z_i^{\epsilon_i} h_i^{-1}$. Consider the covering graph of the corresponding basis rose associated to A. The lift of h_i from its base vertex ends at a vertex v_i. Since $h_i z_i^{\epsilon_i} h_i^{-1} \in A$, uniqueness of path lifting says that the z_i-edge at v_i is a loop edge. These loop edges are distinct for distinct i, since their labels are distinct.

Choose a spanning tree in the covering graph; it contains none of those loop edges. The usual graph basis therefore contains every based loop $t_i z_i t_i^{-1}$, where t_i is the tree path to v_i. The path $h_i t_i^{-1}$ is a loop based at the covering's base vertex and hence represents an element of A. It conjugates $t_i z_i^{\epsilon_i} t_i^{-1}$ to the original given element. The tuple is consequently conjugate to distinct elements of a basis of A, as required. This is the covering graph form of the relevant special case of the Kurosh theorem. □

Theorem 4.3 (Exact recognition on blocks of at most two generators)**.** *Let F_r have a specified basis partitioned into disjoint blocks $X_1, \ldots, X_s$, each of cardinality at most two, and unused generators. Suppose every input conjugacy class is nontrivial and has a cyclically reduced representative over one block. Then the whole tuple is conjugate primitive in F_r if and only if every tuple assigned to a block is conjugate primitive in that block's free group. This is decidable in deterministic polynomial time on compressed words.*

Proof. If every block passes, extend its witnessing automorphism by the identity on all other blocks. The resulting block automorphisms commute and their product sends the whole tuple to distinct generator conjugacy classes. For the converse apply the preceding lemma to each block free factor.

A block of rank one can contain at most one curve, whose exponent must be 1 or -1. A block of rank two can contain at most two curves. Use the known compressed single-class algorithm for one and the pair theorem for two. More curves than the block rank fail by independence in abelianization. Trivial input classes fail. These tests use polynomially many calls on polynomial-size grammars, and all primitivity-discovery calls have the same fixed rank-two bound. □

An automatic version cyclically reduces each word, computes its set of appearing generators from the grammar, and joins generators occurring in the same support. If every nonempty connected component has at most two generators, these components give the required partition. If a component is larger, this particular exact terminal has no conclusion. Failure of a primitive-family test is not a negative handlebody or knot certificate.

4.2 A strict separation from unsigned singleton criteria

Let $\theta(a) = ab$, $\theta(b) = a$, and put $u_n = \theta^n(a)$, $v_n = \theta^n(b)$. This is a basis pair for every n, and it admits a shared grammar of size $O(n)$. For $n \geq 4$, each word contains each generator at least twice. Consequently its unsigned two-by-two incidence matrix has every entry at least two: the private cocore test has no candidate and positive singleton cancellation cannot start. The simultaneous rank-two criterion accepts the pair.

This is a geometric criterion separation as well. A handlebody homeomorphism realizing θ^n takes two standard disjoint longitude curves to a disjoint conjugate primitive pair. Each unsigned geometric cocore intersection count is at least the absolute corresponding homology coordinate, so every relevant core entry is still at least two even if these representatives have extra backtracking. One can then attach a solid-torus orientable interval bundle over a Möbius band to each annulus. Each bundle has just one fully attached boundary and supplies no unit boundary column: that boundary represents the square of its solid-torus generator. Thus each new bundle column has entry two in its own row and zero in the other. Every column total in the two-row unsigned presentation is at least two, so the prior stopping-set theorem excludes every complete positive cancellation in that presentation. The core-side primitive family still gives a genus-two handlebody by the gluing formula: its reduced core genus is zero, the two bundle genera sum to two, and the attachment graph is a tree. An implementation claim about *compressed source geometry* requires a separately authenticated representation of those homeomorphism images. The word grammar by itself is not such a representation.

Two safeguards illustrate why the nonabelian and simultaneous checks remain necessary. The word a^2b^3 has coprime exponent sums but is not primitive: its quotient group has a surjection onto S_3, by sending a to a transposition and b to a three-cycle. A primitive relator in F_2 would instead give a cyclic quotient. In fact the pair (a^2b^3, ab^2) has a unimodular exponent matrix, of determinant one, and still fails the nonabelian test. This last pair is an algebraic control; no source embedding is inferred from those two words. Also, two parallel copies of a primitive boundary curve are individually primitive but do not form a primitive family, since their two homology columns are dependent. No amount of separate single-curve testing repairs that simultaneous defect.

5 From a common basis certificate to the whole-cut census

5.1 The primitive-annulus gluing statement

Theorem 5.1 (Terminal gluing with a primitive core family)**.** *Suppose a finite collection of orientable handlebodies is glued along pairwise disjoint disk and annulus interfaces by orientation-compatible identifications. Suppose every annular interface has exactly one core endpoint and one bundle endpoint, as in an authenticated core/interval-bundle decomposition. Each annulus is assigned to its core endpoint. In each core handlebody H_v, suppose all its incident annular midpoint classes form one conjugate primitive family, of cardinality a_v. Then every connected component of the glued manifold is a handlebody. For a connected component of the attachment graph $G = (V, E)$ its genus is*

$$g(G) = \sum_{v \in V} (g_v - a_v) + |E| - |V| + 1, \tag{2}$$

where $a_v = 0$ at bundle vertices and E contains both kinds of remaining interfaces. Isolated interval bundles contribute their own genera and binary multiplicities.

Proof. By the classical bridge, choose disjoint dual disks for all the assigned annuli at every core vertex, avoiding the other selected annuli and the disk ports. An annulus gluing has the standard relative handle description: connect the two pieces by a one-handle and attach one two-handle around the annulus. Its attaching curve meets the assigned private core one-handle exactly once and meets the other assigned private handles zero times. Cancel this one/two pair. The connecting one-handle remains. Because the disks and interfaces were chosen simultaneously, all these cancellations are compatible.

After doing this for all annuli, vertex v contributes a handlebody of genus $g_v - a_v$, and every graph edge corresponds to a disk gluing or connecting one-handle. Along a spanning tree these operations form boundary connected sums, adding genera. Each of the remaining $|E| - |V| + 1$ edges adds one genus. This proves the topology and the formula. The primitive-family condition also implies $a_v \leq g_v$, so the reduced vertex genera are nonnegative. An isolated orientable interval bundle over a connected bordered surface F is a handlebody of genus $1 - \chi(F)$: retract the base onto a graph and take its regular neighborhood. Each isolated copy contributes independently. □

The formula is already used by the baseline primitive-annulus census. The new theorem application changes how a simultaneous family is certified. No primitive property is required of the annulus on its opposite bundle side: its assigned core dual disk supplies the cancelling handle.

5.2 The Euler identity is a cross-check, not the proof

The checker recomputes graph connectivity, owners, genus deductions, and isolated multiplicities. It also checks

$$\sum_{g} (1 - g) m_g = \chi(\text{core}) + \chi(\text{bundles}) - \chi(\text{interfaces}),$$

where m_g is the claimed number of genus-g components. This can detect certain counting inconsistencies. Exact interface coverage and ownership are enforced separately: annular interfaces have Euler characteristic zero, so the scalar identity alone cannot detect their omission or a changed owner. The handlebody conclusion, however, comes from authenticated geometric cancellations and simultaneous primitive annuli. Euler characteristic alone cannot establish it.

5.3 A local completeness statement and its limit

Corollary 5.2 (Exact terminal on the supported algebraic class)**.** *Assume the geometric prerequisites above, a supported complete core cancellation, and cyclic support components of size at most two in every core's transported annular family. With sufficient resource allowance, the new algebraic stage decides exactly whether those families are primitive. Whenever they are, it constructs a certificate yielding the exact whole-cut handlebody census.*

Proof. Apply the support-block theorem independently in each authenticated core. Its positive certificates give all the families required by Theorem 5.1; replay and the graph calculation then give the census. The exact algebraic decision is independent of expanded word length. □

The converse conclusion "a nonprimitive family means the glued manifold is not a handlebody" is not asserted. Other geometric mechanisms can make the full gluing a handlebody. Likewise, the primitive-family test can be exact on its stated input class while the entire normal-cut classifier is incomplete. The code keeps the outer status INCONCLUSIVE for all these method failures.

6 Algorithm, certificate format, and encoded cost

6.1 Producer stages

The new producer is deterministic once the existing core certificate and source forest have been selected. Its stages are:

1. Reconstruct signed real and passive source words and their owners. Authenticate which real rows came from disk absorption.
2. Compile the complete core schedule into one shared quotient grammar. The row-leaf shortcut maps its cancelled generators to the identity; the column-private case uses reverse triangular equations.
3. Validate that every source terminal lies in the declared component alphabet, *before* free or cyclic reduction can erase it.
4. Cyclically reduce each passive root. Reject the primitive condition if any root is empty. Build the generator-support components from the actual reduced grammar, not from exponent sums.
5. On a rank-one block, require one class with exponent ± 1. On a rank-two block, require at most two classes and run the Euclidean lift test. A larger component is outside the supported discovery class.
6. Combine the disjoint block transformations into one move list, compile its action backwards, and map the source grammar once. Require every final cyclic root to be its specified positive target letter, with distinct targets.
7. Publish source evidence, component coverage, and the common basis certificate. The enclosing operator computes the graph census only after the whole family passes.

Empty annular families are allowed and require an empty target list. An unused core generator is harmless. A nonempty trivial annular class is not primitive. Independent inversion choices can always be incorporated into the automorphism, so the emitted target letters are positive without loss of generality.

6.2 Euclidean matrix convention

Exponent-sum vectors are columns. A Nielsen move $x_i \mapsto x_j^k x_i$ adds k times coordinate i to coordinate j. It does not add coordinate j to coordinate i. Thus for a first column $(p, q)^\mathsf{T}$ with $p, q > 0$, reducing p by $\lfloor p/q \rfloor q$ uses a move whose *target letter is* b and donor is a.

The implementation first changes signs, applies the division version of Euclid, and interchanges the axes if necessary. If the first column has gcd one, it reaches $(1, 0)^\mathsf{T}$. In the two-column case, unimodularity makes the other column $(s, \delta)^\mathsf{T}$ with $\delta = \pm 1$. The shear $b \mapsto a^{-s\delta} b$, followed by an inversion of b when $\delta = -1$, finishes the reduction. Every exponent is stored as one binary integer; a quotient of size 2^b does not cause 2^b repeated elementary steps.

6.3 A small certificate with a broader verifier

The algebraic certificate has fields

```
{
  "schema": "cyclic-basis-nielsen-v1",
  "generators": [sorted positive generator labels],
  "moves": [chronological powered Nielsen moves],
  "targets": [distinct positive labels, one per input root]
}
```

It contains no claimed exponent matrix or claimed reduced word that the checker trusts. The verifier validates the move schemas, label membership, exact integer types, side choices, and nonzero powers. It evaluates their action independently, then performs the terminal word checks. It does not need the producer's support partition or Euclidean choices. In particular, the verifier can certify a supplied primitive family in rank greater than two, even when this producer cannot discover the certificate.

The geometric wrapper records the source packet and one such certificate for each core component. Its source packet lists the forest, real rows, passive attachment/base/core owners, complete cancellation schedule, and surviving generators. Ordered component and slot coverage prevents omitting a difficult annulus or silently moving it to another component.

Schema	Role
`signed-core-annulus-source-v1`	Source-oriented real and passive words; conditional on the enclosing authenticated geometry
`cyclic-basis-nielsen-v1`	One automorphism carrying a cyclic family to distinct generator classes
`signed-primitive-annuli-v1`	Complete family coverage and source transport in every core component
`normal-cut-handlebodies-signed-annuli-v1`	Whole-cut certificate after ordinary core and disk-absorption proofs
`normal-cut-handlebodies-signed-annuli-v2`	Same terminal after replay of the existing top 2–3 reduction

Table 2: New schema boundaries. Older schemas retain their original meanings.

6.4 Independent replay and the shared trust base

The checker modules do not import the producer, its Euclidean selector, its support decomposition, or its source compiler. The signed source checker reconstructs the geometry in a separate implementation. The quotient checker validates the cancellation order by deletion timestamps and source incidences, then evaluates the signed donor recursion independently. The basis checker uses suffix-image layers and a separate left-to-right binary-power loop.

For a column-private schedule, if $\tau(r)$ is the row's deletion time and $\pi(g)$ the pivot time of a cancelled generator, every source occurrence of g in row r must satisfy $\tau(r) \leq \pi(g)$. Equality is possible only in its own pivot row, where the occurrence count must be exactly one. This gives one pass over source incidences, plus one pass over donor words, instead of rescanning all rows for each pivot.

Both production and replay use the baseline `WordArena` for exact compressed reduction. Hence "independent" does not mean two independent implementations of every theorem in compressed-word theory. The independent literal audit described below supplies a separate reduction and Whitehead oracle on bounded inputs. The existing source and word kernels remain part of the software trust base; the article's mathematical proofs do not turn Python code into a formally verified development.

6.5 Polynomial encoded cost

Proposition 6.1 (Cost of the local algebraic stage)**.** *With integer labels encoded in binary, signed source transport, support decomposition, exact rank-at-most-two block discovery, and verification of the emitted certificate have deterministic polynomial bit complexity in the authenticated source encoding. The degree does not depend on the overall core genus. For a separately supplied Nielsen certificate, the verification bound is polynomial in the source and certificate encodings.*

Proof. The signed source theorem gives $O(L + r + p + q)$ raw grammar productions. Inverse roots and binary products preserve a linear structural bound. All raw lengths have polynomially many bits. Deterministic compressed cyclic reduction gives grammars of polynomial encoded size.

For each reduced grammar node, compute support by set union and exponent sums by integer addition. Even dense arrays cost only a polynomial factor in r. Connected components and membership checks are polynomial. All nontrivial matrix calculations occur in dimension at most two. Euclidean division has polynomial bit complexity, and the quotients and resulting move list have polynomial total encoding size. This is uniform over polynomially many separate blocks.

Backward replay adds $O(r + B)$ raw productions before applying the source grammar once. The deterministic compressed reduction theorem therefore bounds the final checks polynomially. The independent checker stores $O(rm)$ suffix-image references for m moves; this is polynomial, although less economical than the producer's mutable image map. Integer lengths, reference sorting, and component bookkeeping are charged in the encoded bound. No step iterates over the expanded passive word length. □

The implementation's resource limits are operational guards, not asymptotic claims: by default the new word arenas allow 10^6 nodes and 10^7 charged abstract operations. Python dictionary access is not being asserted to have deterministic worst-case constant time against adversarial labels. Ordered maps or sorted arrays can realize the polynomial theoretical bound.

The current source adapters also scan one-handles separately for each core component, which can cost quadratic structural work. The measured wrapper still has the source-query, forest, normal-surface, and geometric verification costs of its existing stages.

The uniformity in rank uses the compressed free/cyclic reduction and composition-system results, not just a fixed-group automorphism word problem: see Schleimer's Theorem 3.3, Corollary 3.6, and Theorem 7.1 [10]. Renumber the occurring letters; their number is bounded by the input's terminal productions. The algorithms operate on symbolic equality, involution, and compressed prefix comparison, so label bits can be charged without introducing a degree depending on the ambient rank. Initial cyclic reduction may involve the full alphabet; only the subsequent primitivity discovery is blockwise rank two.

7 Native evidence and reproducible experiments

7.1 An actual signed obstruction in the discovery bank

The first new case is `finite_figureEight/vertex71`, from the standard vertex surfaces of report 57's discovery corpus at the pinned revision. Its triangulation has ten tetrahedra and its normal coordinates represent thirty disks. The embedded fixture includes the exact gluing permutations, all coordinates, and the SHA-256 of the source bank. Thus reproduction does not depend on a surface enumeration library returning the same ordering.

The baseline authenticates a complete augmented core but refuses the remaining annuli with reason `annular_core_attachment`. On the relevant core component, label the surviving generators $a = 29$, $b = 39$ and the disk-absorption generator $z = 55$. The two authenticated passive source words are

$$za^{-1}b^{-1}, \qquad z^{-1}b^{-1}a^{-1}b.$$

The genuine absorbed disk supplies the real row z, so the core quotient sets $z = 1$. The residual conjugacy classes are therefore represented by

$$u = a^{-1}b^{-1}, \qquad v = a^{-1}. \tag{3}$$

Their exponent-sum matrix is

$$A = \begin{pmatrix} -1 & -1 \\ -1 & 0 \end{pmatrix}, \qquad \det A = -1.$$

The emitted chronological moves invert a, invert b, send $b \mapsto a^{-1}b$, and interchange a, b. Their effect on the two cyclic representatives is visible without a large computation:

$$(a^{-1}b^{-1}, a^{-1}) \longmapsto (ab^{-1}, a) \longmapsto (ab, a) \longmapsto (b, a) \longmapsto (a, b).$$

The certificate checks the unabridged source paths, rather than accepting these displayed shortened words as evidence.

There are two core handlebodies of genus two. One has no pending annuli; the other meets an annulus-base interval bundle, a genus-one handlebody, along its two boundary annuli. For the connected attached part, Equation (2) gives

$$(2 - 2) + 1 + (2 - 2 + 1) = 2.$$

The isolated core contributes the other genus-two component. An independent Regina construction cuts the actual triangulation into 297 tetrahedra and recognizes exactly two genus-two handlebodies.

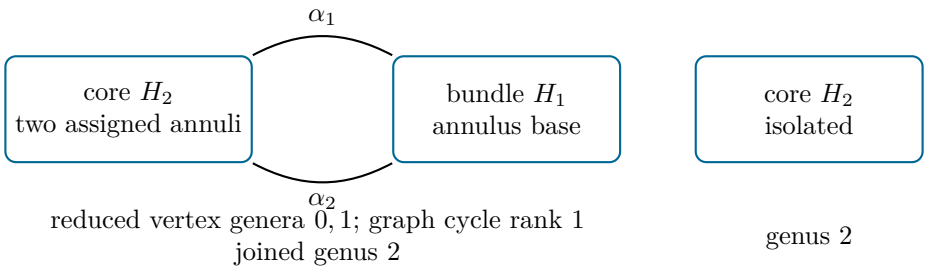


Figure 1: The first native gain's attachment graph. Both parallel edges must be included in the simultaneous primitive family.

This is a cut in a figure-eight exterior. A handlebody census of a chosen cut is not an unknot verdict. A nontrivial knot can admit a cut into handlebodies; the hierarchy's boundary patterns and the surface's role still matter. The example validates the local source-to-census inference and its strict success-set improvement.

7.2 Further native gains, retained refusals, and the audit denominator

The extended report-57 scan contains 1,374 standard vertex surfaces, excluding its Fibonacci and connected-sum records. The baseline completes 1,205 and refuses 169. The recorded baseline refusal reasons are 52 raw cancellation stalls, 101 boundary-kernel stalls, five raw ball-cancellation obstructions, and eleven annular attachments. Inspecting the refused cases under the tested source, maximum-passage, and reverse-ID forests finds 52 with a complete augmented core and pending annuli. Exactly five of those families pass the signed common-basis test. All five now pass the complete source-authenticated wrapper with its production forest policy.

Case	t	W	Source letters	Word nodes	Moves	Regina cut t
E71	10	30	45	16	4	297
I134	13	37	56	24	2	377
I136	13	37	59	18	3	386
I205	13	38	57	19	3	398
I290	13	38	57	21	2	398

Table 3: Five new native gains. E denotes `finite_figureEight`; I denotes `finite_figureEight_interior`; numbers are the frozen standard-vertex indices. Every result is two genus-two handlebodies, with independent whole-cut replay and Regina 7.4 agreement.

E71 uses the ordinary signed schema. I134, I136, I205, and I290 first use the established top 2–3 reduction and the second signed schema. The five complete certificate files have canonical sizes 144,704; 199,262; 160,239; 198,738; and 198,726 bytes respectively. Most evidence is the existing source geometry, rather than the short Nielsen move list.

In one recorded run, whole-cut production took between 0.053 and 0.105 seconds on these five cases, and full independent replay between 0.075 and 0.118 seconds. These are single-run observations on a shared execution host, not an isolated timing comparison or a claimed multiplicative speed-up. The baseline timings measure refusals; the new timings measure completed certificates. Source encodings and exact operation counts are the more reproducible performance evidence.

Two other exploratory banks produced no new opportunities: all 660 tested Fibonacci

vertex/scaled cuts and all 1,700 deterministic seed relabelings were already completed by the baseline. The broader 1,374-case discovery search took 562.07 seconds and included a separate literal Whitehead search used to investigate refused states. That exploration time is not the cost of the new compressed production algorithm.

The discovery procedure also retains a negative control whose transported annular word is x^5y^2. It has coprime exponent sums, but it is not primitive: the one-relator quotient maps onto $C_5 * C_2$, which is nonabelian, whereas quotienting F_2 by a primitive element gives an infinite cyclic group. The new method still refuses this cut. This is a failure of the primitive terminal, not a proof about the topology of the entire cut.

No recognition-rate claim is based on pooling the historical banks, relabelings, and new fixtures as independent random observations. Relabelings test invariance and exposure, while related normal surfaces can be strongly correlated. The frozen JSON files record each exploration's denominator, reason counts, and positive cases separately.

All five gains were also replayed with the public default `boundary_method='masks'`, in addition to the recorded moments-based run. Thus the new terminal also works with the default boundary setting.

7.3 An independent algebraic oracle

The independent audit enumerates all sixteen rank-two type-II Whitehead maps and repeatedly decreases the *total cyclic length* of a tuple. It accepts exactly when a length-minimal tuple consists of distinct one-letter classes. This uses the classical Whitehead minimization principle; the audit's literal free/cyclic reductions and substitutions do not import the producer, its determinant test, or `WordArena`. General explicit-input minimization already has fully polynomial algorithms in length and rank [9]; the tiny exhaustive oracle is used for independence, not as a new efficient algorithm.

Decision family	Inputs	Positive
Nonempty cyclically reduced rank-two words, length at most six	1,104	196
Ordered pairs, nonempty and total cyclic length at most six	6,576	648
Random unstructured singleton/pair inputs	400	41
Random automorphic images with independent conjugations/inversions	397	397
Named controls	9	2
Total	8,486	1,284

Table 4: Independent algebraic audit, seed 20261012. There were zero decision discrepancies and every positive certificate replayed correctly.

Additionally, 396 supplied Nielsen sequences in ranks two through five matched explicit forward substitution on 4,008 unreduced outputs, with 4,008 inverse roundtrips. The largest literal output had 36,687 letters. The sequences cover both multiplication sides, both exponent signs, inversions, swaps, and powers of absolute value greater than one.

All 71 malformed certificate mutations were rejected. Among 24 syntactically valid alterations, twenty were invalid proofs and rejected; four remained valid proofs and were accepted, exactly as the independent literal replay predicted. This distinction avoids the incorrect test rule that every changed certificate must fail. Three attempted random positive generations and six compiler generations exceeded the *literal oracle's generation caps before testing*; they are reported as such rather than counted as production failures. The audit ran in 6.514 seconds

on the recorded environment. Finite tests support implementation confidence; they do not prove the asymptotic theorem.

7.4 Compressed stress tests

The benchmark uses four reproducible families. For the Fibonacci family, $(u_n, v_n) = (\theta^n(a), \theta^n(b))$ with $\theta(a) = ab$, $\theta(b) = a$; its total expanded length is F_{n+3} when $F_0 = 0, F_1 = 1$. The shear family is $(a, a^{2^b}b)$, with total length $2^b + 2$. Independent support blocks use disjoint copies of the depth-eight Fibonacci pair. The triangular source family has relations $g_i = g_{i+1}^2$, ends at b, and passive word ag_0, so its final length is $2^d + 1$.

These are explicitly algebraic inputs. The triangular packets test the quotient compiler, not source geometry, and no knot diagram or embedded normal surface is inferred from them. Every positive result is independently replayed in a freshly built word arena.

Family	Parameter	bits(N)	Input DAG	Final DAG	Produce ms	Verify ms
Fibonacci pair	128	90	130	773	2.57	2.23
Fibonacci pair	2,048	1,423	2,050	12,293	41.63	34.69
Binary shear pair	128	129	131	264	1.32	0.99
Binary shear pair	2,048	2,049	2,051	4,104	17.54	12.65
Triangular transport	128	129	131	265	0.72	0.74
Triangular transport	2,048	2,049	2,051	4,105	12.16	12.70
Independent blocks	16	11	160	848	2.82	2.25
Independent blocks	64	13	640	3,392	11.85	10.75

Table 5: Selected compressed benchmarks; all positive results independently verified. Parameters are depth n, exponent-bit parameter b, transport depth d, and number of blocks, respectively. N is total expanded root length, with $\text{bits}(N) = \lfloor \log_2 N \rfloor + 1$. No expanded words were materialized. Full measurements and code hashes are in the JSON.

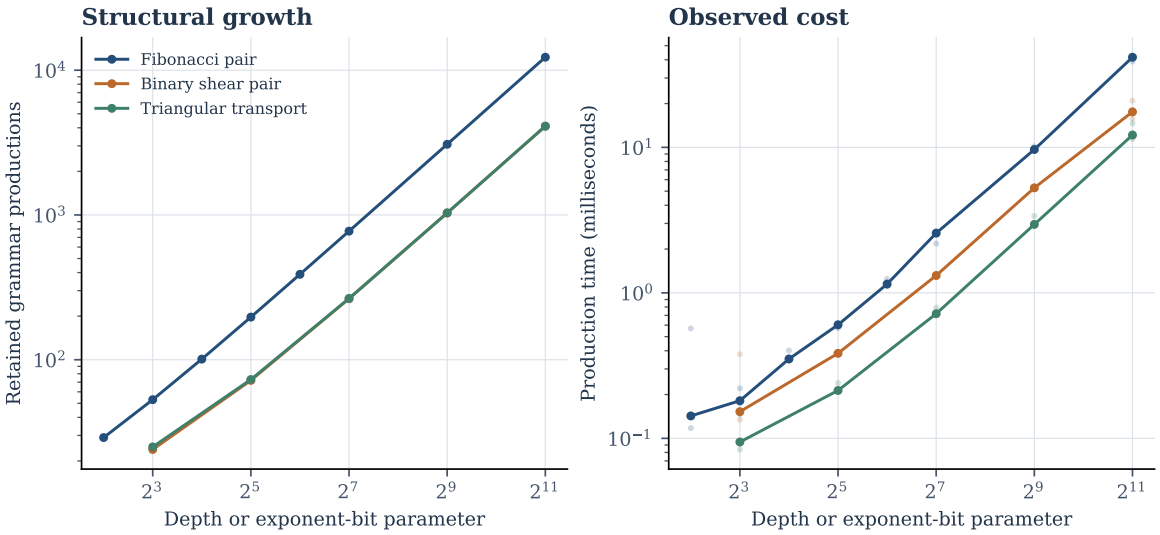


Figure 2: Recorded grammar sizes and elapsed production time for three compressed families. The horizontal parameter controls exponentially large expanded words. Straight-line growth of stored grammar sizes on these examples illustrates the proved sharing mechanism; the finite timing curves do not establish an asymptotic exponent.

Times are medians of three runs. Production time excludes construction of the input grammar, and verification time excludes its reconstruction; separate construction timings are stored in the JSON. Grammar counts include arena nodes retained after processing, while the input column counts only reachable source productions. The theoretical statements concern all valid compressed inputs in the supported class, not an extrapolation of these particularly structured families.

7.5 Repository regression and reproducibility limits

All 1,741 collected repository tests were exercised successfully: 1,733 passed in the final full run, together with 1,750 passing subtests; eight tests were blocked at setup by a missing historical synthesis fixture and passed in a separate 0.99-second run after its exact restoration. The full run took 506.68 seconds and had no assertion failures. The focused suite contains eighteen new tests covering algebra, signed source reconstruction, and whole-cut integration.

The frozen logs preserve the setup failures rather than presenting them as one uninterrupted green invocation. An earlier run also encountered workspace exhaustion, a missing historical commit, and two stale refusal expectations. Those two expectations already failed against the untouched baseline because adaptive forests now complete the examples. They are pinned to `lower_forest='source'` in the patch, preserving the historical obstruction each test was intended to exercise. No production behavior was changed to satisfy those assertions.

The included snapshot supports the eighteen focused tests, the native fixtures, and the algebraic audits without a full repository checkout. Replaying the entire upstream suite requires its original report and synthesis fixtures and one historical commit; the README records those dependencies. Regina is needed only for the optional expanded-cut cross-check. These dependency boundaries are separate from the mathematical and certificate correctness claims.

8 What would suffice for the global quasi-polynomial bound?

8.1 A quantitative transfer statement

The following elementary consequence of a hierarchy iteration bound is a useful specification for the remaining programme. It isolates the exact quantities a local performance result must feed.

Proposition 8.1 (Conditional hierarchy accounting)**.** *Consider a* complete and correct *hierarchy recognizer on diagrams of $n \geq 2$ crossings, with $L(n) \geq 1$, $g(n) \geq 0$, and $B(n) \geq 2$. Suppose it executes at most $L(n)(g(n)+1)^{L(n)}$ iterations, and each iteration, including discovery, repairs, and terminal work, takes at most $B(n)^d$ bit operations for a fixed constant d. Assume its combined preprocessing and postprocessing also take at most $B(n)^d$ bit operations. Then*

$$\log_2 T(n) \leq \log_2 L(n) + L(n) \log_2(g(n)+1) + d \log_2 B(n) + O(1). \tag{4}$$

In particular, if $L(n) = O((\log n)^2)$, $g(n) = n^{O(1)}$, and $\log B(n) = O((\log n)^3)$, the recognizer has the bound in Equation (1). Polynomially sized states and polynomial local work are a stronger sufficient version of the last hypothesis.

Proof. Multiply the number of iterations by their worst-case bit cost. The preprocessing and postprocessing cost at most one extra iteration bound, so the total is at most $2L(n)(g(n)+1)^{L(n)}B(n)^d$. Take base-two logarithms. The stated growth assumptions make every term on

the right $O((\log n)^3)$. This counts repeated repairs and discarded work only because they were expressly included in the iteration and local-cost hypotheses. Ignoring them would invalidate the bound. □

The proposition is a conditional accounting lemma, not a derivation of the missing hypotheses. It is compatible with Lackenby's stated iteration analysis and announced speed-up mechanism, but does not claim to reconstruct that unpublished quantitative argument. The polynomial terminal in this article helps satisfy one local-cost obligation when its structural preconditions occur.

8.2 Why certificates alone do not supply the search bound

A short certificate can be easy to verify and hard to discover. Backward Nielsen replay controls the size of the action of a *supplied* sequence; it does not control how many transformations a search tries. Likewise, a primitive-family certificate proves the existence of a disk system without constructing its full boundary embedding for use by a later hierarchy.

In rank two, the abelianization map resolves the discovery problem because its outer kernel is trivial. In rank at least three this simplification fails even for a very elementary family. In $F(a, b, c)$ the automorphism

$$a \longmapsto a[b, c], \qquad b \longmapsto b, \qquad c \longmapsto c$$

has identity abelianization. The tuple $(a[b, c], b, c)$ is a basis, but the identity lift of its homology matrix leaves a cyclic word of length five. Thus the rank-two "choose any lift" argument cannot be extended by merely enlarging the matrix. Nontrivial IA automorphisms remain to be found.

Similarly, compressed Whitehead *minimality testing* and compressed Whitehead *minimization* are distinct tasks. The former asks whether a shortening move exists at the current state; the latter follows enough moves to reach an orbit minimum. The number of strict decreases can be governed by the enormous expanded length. Kapovich explicitly distinguishes these issues for compressed words [4]. A sequence of polynomial steps is not a polynomial algorithm without a polynomial step count.

8.3 Marked geometry is the next interface

The current output records a topology type and a source-valid primitive family. A complete hierarchy may need a description of where every old pattern edge, disk interface, and new meridian lies after the basis change. An automorphism of the free group records none of those intersections by itself. Handlebody homeomorphisms can act identically on the free group up to inner automorphism while affecting boundary data [3]. The marked problem therefore has strictly more information to carry.

A productive next representation would maintain a compressed boundary diagram with a rotation system, local source charts, and shared path expressions. A basis certificate would be accompanied by a checked realization of the elementary moves on that diagram. Such a representation must prove that its encoded size remains controlled through repeated composition, including substitutions caused by geometric repairs. This is an explicit research obligation, not an automatic consequence of the SLP word bound.

9 Further research questions and proposed approaches

The questions below are ordered by the value of a concrete answer to the existing implementation. Each states an object to construct or a bound to prove, and an experiment or obstruction that can make progress measurable. They are proposals, not asserted theorems.

9.1 From terminal existence to usable marked geometry

Research question 1 (Compressed realization of the dual disks)**.** Given authenticated annular source curves and a polynomial-size common Nielsen certificate, can one construct a comparably sized certificate of the simultaneous dual disks, together with their intersections with a prescribed boundary pattern?

A first restricted target is one rank-two block in a core with only disk ports. Maintain the boundary diagram of each Nielsen move as a local band-slide operation, then share repeated bands using a grammar. A proof must bound both the combinatorial chart count and the bit lengths of repeated traversals. The decisive test is whether the resulting object can serve as input to the next authenticated normal-cut or pattern operation.

Research question 2 (Primitive families relative to omitted annular ports)**.** What additional certificate characterizes a primitive subfamily whose dual disks must avoid a specified collection of unselected annular ports?

The present theorem only removes disk ports freely. Relative cyclic data could be encoded by a free-factor system or a peripheral multiword, but the algebraic condition must be proved equivalent to the required embedded relative disks. An initial search should look for pairs of boundary embeddings with identical unmarked free-group data but different relative disk behavior. Such examples would identify exactly which marking data cannot be discarded.

9.2 Larger algebraic blocks without expanded descent

Research question 3 (Homology normalization followed by sparse IA correction)**.** After using integral row operations to normalize the homology of an annular family, can the remaining IA obstruction be eliminated efficiently when it is supported on a sparse interaction graph?

One proposed procedure checks that the homology columns span a direct summand, completes them to an integral basis, and lifts the resulting unimodular matrix to powered Nielsen transformations. It then studies the residual words with unit homology. The rank-three example $a[b, c]$ shows what this second phase must detect. A promising precise restriction is that each residual conjugation or commutator involves a bounded-size separator of the interaction graph. The proof target is a bound on the total *binary size* of corrections, not merely on their individual evaluation cost.

Research question 4 (Support decomposition after certified basis changes)**.** Can one find a short sequence of basis changes that separates a given family into blocks of rank at most two, or certify a useful obstruction to such a sequence under a specified budget?

The current general verifier can already check the concatenation of a supplied preliminary basis change and the exact block certificates. This permits exploration without enlarging the trust base. A search could try weighted Whitehead cuts or forest-induced changes, minimizing the largest support component before minimizing length. Completeness must not be claimed from this heuristic objective: a primitive family always admits some basis with singleton supports, so unrestricted optimization already contains the difficult discovery problem.

Research question 5 (Polynomial compressed minimization for native geometric families)**.** Does the annular origin of the multiwords force a quantitative restriction on Whitehead descent that arbitrary compressed free-group words lack?

The useful hypothesis should refer to the authenticated source embedding, the handle structure's uniform type, or the topology of the interval-bundle bases. Bounds expressed only in the SLP size are the general compressed problem again. A first experiment should measure strict decreases, plateau moves, and peak grammar size on the remaining native annular refusals, and retain the full source certificates for any apparent structural pattern.

9.3 Stronger source transport and terminal coverage

Research question 6 (Signed transport beyond no-fill core schedules)**.** Can arbitrary certified positive-fill or geometric disk-slide schedules be compiled into a polynomial shared signed peripheral grammar without rebuilding every passive root at every step?

The column-private theorem works because the real donors remain original source words. For general schedules one should record donor expressions as shared definitions and maintain a persistent substitution circuit. The main obstruction is composition: blindly substituting a new circuit into every old circuit may destroy the desired bound. A successful theorem must control the combined DAG and its integer lengths, as the backward Nielsen argument does for automorphisms.

Research question 7 (Mixed core-side and bundle-side primitive families)**.** Can compatible partial families on both sides be combined with the signed terminal without an exponential assignment search?

The baseline joint boundary kernel already handles substantial mixed unsigned behavior. A new result must therefore use information that it does not already capture. One possibility is a compatibility graph whose vertices are *certified simultaneous* primitive families, with edges encoding which interfaces can share a disk system. Whether this graph has a matroidal or bounded-width structure must be proved from the geometry; choosing each annulus's preferred side independently is unsound.

Research question 8 (A broader exact local terminal)**.** Which native handlebody gluings fail primitivity on both sides but admit a different short geometric certificate?

The retained x^5y^2 case is a useful starting point because it defeats a specific primitive inference while keeping its provenance fixed. The next step is to inspect a Regina disk system in the actual cut and translate its mechanism into an elementary source-checkable move. Algebraic freeness, homology, or a successful external recognition call alone should not become a certificate schema.

9.4 Quantitative hierarchy construction

Research question 9 (A constructive multisurface selection bound)**.** Can the proposed multisurface stage output all its normal coordinates, compatibility data, and useful homology classes within a bound compatible with $L = O((\log n)^2)$ and $g = n^{O(1)}$?

An implementation milestone should separate finding a candidate multisurface, authenticating disjointness and orientation, and proving that the selected state decreases the controlling

potential. The cost of enumerating every fundamental surface is not an acceptable substitute for the desired asymptotic statement. Record the number of source handles and coordinate bits at every stage to expose where a supposedly short hierarchy acquires an expensive state.

Research question 10 (Cheeger-region and Heegaard operations with certificates)**.** What finite combinatorial objects encode the announced region and weak reduction operations, and what is their worst-case discovery cost?

The first deliverable should be a precise input/output contract, including the boundary pattern inherited by each output piece. Every use of a geometric existence theorem must identify an effective search or a short verifiable witness with a proved discovery bound. The signed annulus terminal may close some output pieces, but it cannot stand in for these region-selection procedures.

Research question 11 (Amortizing hierarchy repairs and discarded suffixes)**.** Can a persistent history reuse certified compressed geometry after a violating disk repairs an earlier surface, while charging all discarded work to a quasi-polynomial potential?

An explicit target is a potential with at most $2^{O((\log n)^3)}$ decreases and a polynomial-size update certificate. Reused states need stable source identities so that a changed prefix cannot silently reuse an invalid suffix certificate. A useful first experiment compares complete histories of the same inputs with and without reuse, reporting total work, not only the last successful hierarchy.

Research question 12 (A complete marked terminal and a global cost ledger)**.** Can every terminal state required by the hierarchy algorithm be reduced to an exact pattern-essentiality or violating-disk certificate, with a uniform bound on the encoding and work?

This is the point where a handlebody census must connect to the boundary pattern rather than discard it. Build a ledger containing L, g, maximum state bits, primitive-search work, pattern work, and repair counts. A proof of the inequalities in Proposition 8.1, with every term tied to an executable operation, would establish the intended asymptotic milestone. Until then the local improvements should retain their precise names and contracts.

10 Reproduction and integration

The accompanying archive contains this article in LaTeX and PDF, bibliographic sources, figures and their generating script, the integration patch against the pinned repository revision, a runnable snapshot of the needed Python package, native fixtures and full certificates, the independent algebra audit, compression benchmarks, and frozen test logs. The manifest records hashes so that measured code can be compared with the delivered code. No external repository branch was changed by this work.

The top-level README gives commands for applying the patch or running the included snapshot. The principal native invocation is:

```
normal_cut_handlebodies(
    triangulation, coordinates,
    boundary_method='moments',
    annulus_method='signed_rank_two',
    record_certificate=True,
)
```

The returned certificate is separately checked by the function `verify_normal_cut_handlebodies`, with the triangulation, coordinates, and certificate as its three arguments. The optional Regina cross-check constructs the expanded cut only for these small validation fixtures; production and certificate replay do not depend on that expansion.

For acceptance into the repository, preserve the source bank identifiers, new negative control, malformed-certificate cases, and the exact distinction between source geometry and algebraic benchmarks. The default can be reconsidered after a wider workload study. The present evidence establishes a strict local success-set extension and controlled compressed algebra; it does not justify a universal wall-time speed-up claim on all inputs.

A Mathematical counterexamples as regression specifications

Input family	Tempting insufficient test	Correct conclusion
(a, a)	Each word is primitive	Not a primitive family; repeated homology column
(a, ab^2)	Each word is primitive	Not a primitive pair; determinant two
(a^2b^3, ab^2)	Determinant one	Not primitive; the first quotient has a nonabelian S_3 quotient
(bab^{-1}, aba^{-1})	Based representatives must literally be a basis	Primitive as cyclic classes; independent conjugators are permitted
$(a[b, c], b, c)$	Identity homology lift must suffice in every rank	Primitive family; the rank-two lift argument does not extend to rank three
$(\theta^n(a), \theta^n(b))$, $n \geq 4$	No unsigned singleton means no primitive family	A basis pair with every unsigned entry at least two
Source word cac^{-1} with c outside the declared alphabet	Cyclic reduction erases the undeclared symbol	Invalid source alphabet; reject before reduction
True primitive family with an omitted pending annulus	Certify only the easy subfamily	Insufficient geometric evidence; require full coverage

B Review checklist tied to proved hypotheses

This checklist states the dependencies of the inference, rather than additional user approvals or a replacement for the tests.

1. The source geometry and normal coordinates are independently authenticated; the raw words are not treated as embedded curves by fiat.
2. Every signed crossing uses ordered source flags, including loop one-handles, and every disk-absorption row is reconstructed.

3. The core proof establishes a geometric handlebody with the marking used by the quotient grammar. Any top reduction is replayed first.
4. The supported complete schedule is row-leaf or column-private; arbitrary capped fill is not silently routed into the sharp source theorem.
5. Every passive root belongs to the correct surviving component alphabet; all pending annuli occur exactly once in component coverage.
6. A single supplied automorphism sends the whole component family to distinct generator conjugacy classes, checked by exact word operations.
7. Genus deductions, owners, graph components, and isolated multiplicities are recomputed, with the independent Euler identity retained.
8. Unsupported cases and budget exhaustion remain method refusals; an algebraic negative does not become a knot or manifold negative certificate.

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